Maclaurin Series & Convergence of Series

ClipSAT Printable Practice Suite (clipsat.org) • Topic 5.03

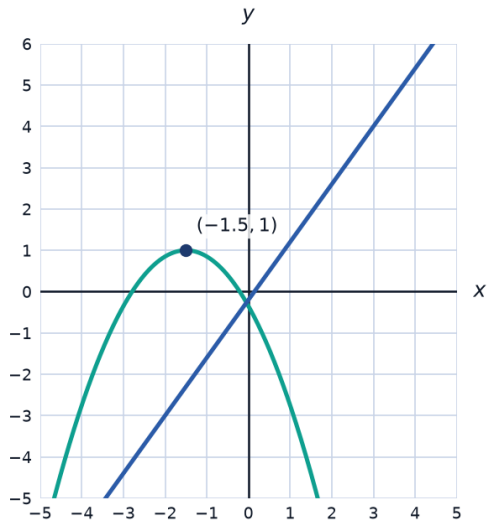
Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Find the coefficient of x^2 in the Maclaurin series of e^{3x}.

2 [Foundational] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 2 · Medium Determine whether the series $\sum_{n=1}^{\infty} \frac{(-1)^{n+1}}{\sqrt{n}}$ converges absolutely, converges conditionally, or diverges.



3 [Foundational] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · Foundations Use the Divergence Test to determine whether $\sum_{n=1}^{\infty} \cos\left(\frac{1}{n}\right)$ converges or diverges.

4 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Level 2 · Medium Find the third-degree Maclaurin polynomial $P_3(x)$ for $f(x) = \cos x$.

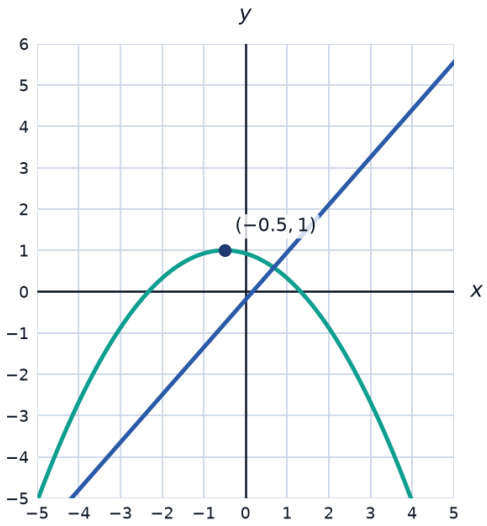
5 [Foundational] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Convergence Tests & Power Series · Practice 4] If the governing relation for this system is $g(t) = 6t + 6$, evaluate $g(10)$ and find the inverse input t when $g(t) = 72$.

© ClipSAT — clipsat.org | All Rights Reserved

6 [Foundational] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Find the first three non-zero terms of the Maclaurin series for e^{2x}. State the range of validity.



7 [Foundational] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

In an arithmetic progression, the 3rd term is 14 and the 7th term is 34. (a) Find the first term a and the common difference d. (b) Find the sum of the first 25 terms.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Level 3 · Advanced Let $f(x)$ be a function with $f(2) = 3$, $f'(2) = -1$, $f''(2) = 4$, and $f'''(2) = -12$. Write the third-degree Taylor polynomial $P_3(x)$ for f centered at $x = 2$, and use it to approximate $f(2.1)$.

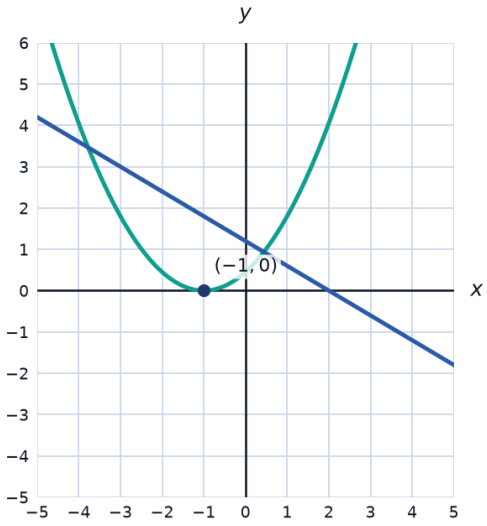
9 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

Level 1 · Foundations Find the exact sum of the infinite geometric series: $\sum_{n=1}^{\infty} 3\left(-\frac{2}{5}\right)^n$

© ClipSAT — clipsat.org | All Rights Reserved

10 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

[Sequences & Series · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.



11 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

[Series & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

12 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

[Sequences, Series, Induction & the Binomial Theorem · Practice 3] If the governing relation for this system is $g(t) = 5t + 5$, evaluate $g(8)$ and find the inverse input t when $g(t) = 50$.

13 [Intermediate] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7)

Level 2 · MediumFind the minimum number of subintervals n required to guarantee that the Trapezoidal approximation of $\int_1^3 \frac{1}{x}\,dx$ has an absolute error less than 10^{-4}.

14 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

Find the set of values of the constant k for which the equation $kx^2 + 4x + (k - 3) = 0$ has real roots.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

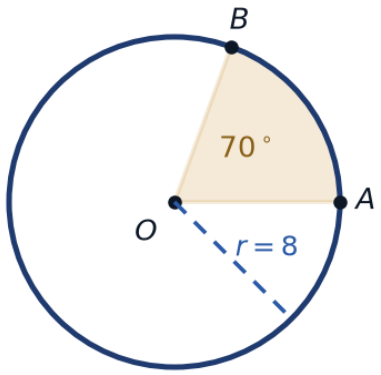
15 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 4.02 The Poisson Distribution

Find the equation of the normal to the curve $y = \frac{4}{\sqrt{x}} + 2x$ at the point where $x = 4$.

© ClipSAT — clipsat.org | All Rights Reserved

16 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 5.01 Further Integration Techniques

Level 2 · Medium Find the total arc length of the parametric curve $x = \cos^3 t$, $y = \sin^3 t$ (the astroid) for $0 \le t \le \frac{\pi}{2}$.



17 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 5.02 Differential Equations

Solve the equation $2\cos 2x = 1 + \sin x$ for $0 \le x \le 2\pi$ radians, giving answers to 3 significant figures.

18 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Level 3 · Advanced Find the exact surface area generated by revolving the line segment $y = 2x$ on $[0, 3]$ about the x-axis.

19 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Matrices (AI HL)

Level 2 · MediumSolve the transcendental equation analytically for x in exact form using natural logarithms: $3^{2x+1} = 5^{x-2}$.

20 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A block of mass $8\ \text{kg}$ rests in limiting equilibrium on a rough plane inclined at 30° to the horizontal. Taking $g = 9.8\ \text{m/s}^2$: (a) calculate the normal reaction R, (b) find the coefficient of friction μ between the block and the plane.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

© ClipSAT — clipsat.org | All Rights Reserved

21 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the general formula for the Maclaurin series expansion of a function $f(x)$?

a. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!}x^n = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \ldots$

b. $f(x) = \sum_{n=0}^{\infty} f^{(n)}(0)x^n$

c. $f(x) = \sum_{n=1}^{\infty} \frac{f(n)}{n!}x^n$

d. $f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x)}{n!}x^n$

22 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first three non-zero terms of the Maclaurin expansion of $f(x) = \cos(2x)$.

a. $1 - 2x^2 + \frac{2}{3}x^4$

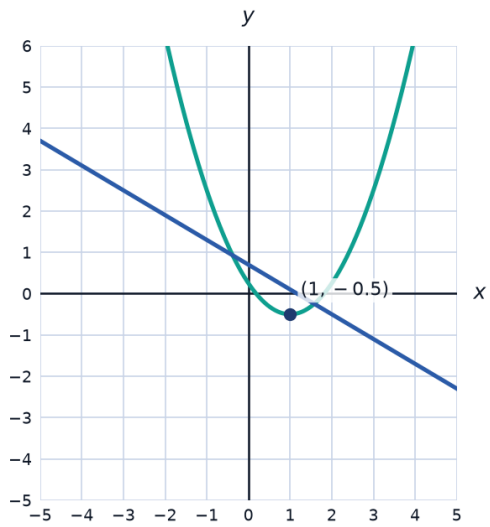
b. $1 - \frac{1}{2}x^2 + \frac{1}{24}x^4$

c. $1 - 4x^2 + 16x^4$

d. $2x - \frac{4}{3}x^3 + \frac{4}{15}x^5$

23 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What is the coefficient of x^3 in the Maclaurin series of e^{-3x}?



a. $-\frac{9}{2}$

b. $\frac{9}{2}$

c. -9

d. -27

© ClipSAT — clipsat.org | All Rights Reserved

24 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the Maclaurin series for $\ln(1+3x)$ up to the x^3 term.

a. $3x - \frac{9}{2}x^2 + 9x^3$

b. $3x + \frac{9}{2}x^2 + 9x^3$

c. $x - \frac{1}{2}x^2 + \frac{1}{3}x^3$

d. $3x - 9x^2 + 27x^3$

25 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Evaluate $\lim_{x\to 0} \frac{e^x - 1 - x}{x^2}$ using Maclaurin series.

a. $\frac{1}{2}$

b. 1

c. 0

d. ∞

26 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Find the first two non-zero terms of the series for $\frac{1}{1+x^2}$.

a. $1 - x^2$

b. $1 + x^2$

c. $1 - x$

d. $x - x^3$

27 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

What does the Ratio Test state for a positive series $\sum a_n$ when $L = \lim_{n\to\infty} \frac{a_{n+1}}{a_n}$?

a. Converges if $L < 1$, diverges if $L > 1$, test is inconclusive if $L = 1$

b. Converges if $L = 1$, diverges otherwise

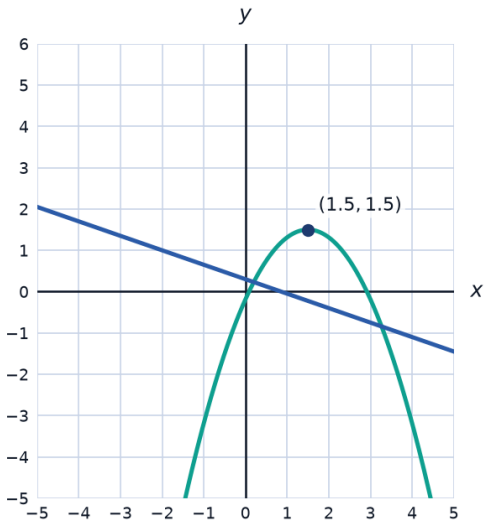
c. Diverges if $L < 1$, converges if $L > 1$

d. Always inconclusive for power series

© ClipSAT — clipsat.org | All Rights Reserved

28 [Foundational] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Determine the convergence of $\sum_{n=1}^{\infty} \frac{n}{3^n}$ using the Ratio Test.



a. Converges, because $L = \lim \frac{n+1}{3^{n+1}} \cdot \frac{3^n}{n} = \frac{1}{3} < 1$

b. Diverges, because $L = 3 > 1$

c. Inconclusive, because $L = 1$

d. Diverges, because the terms do not approach 0

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

29 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the limit L in the Ratio Test for the harmonic series $\sum_{n=1}^{\infty} \frac{1}{n}$?

a. 1 (test is inconclusive)

b. 0

c. ∞

d. 1/2

30 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Find the radius of convergence R for the power series $\sum_{n=1}^{\infty} \frac{(2x)^n}{n}$.

a. $R = \frac{1}{2}$

b. $R = 2$

c. $R = 1$

d. $R = \infty$

© ClipSAT — clipsat.org | All Rights Reserved

31 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

What is the radius of convergence of the Maclaurin series for e^x?

a. $R = \infty$

b. $R = 1$

c. $R = e$

d. $R = 0$

32 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

For $\sum_{n=0}^{\infty} x^n$, what is the exact interval of convergence?

a. $(-1, 1)$

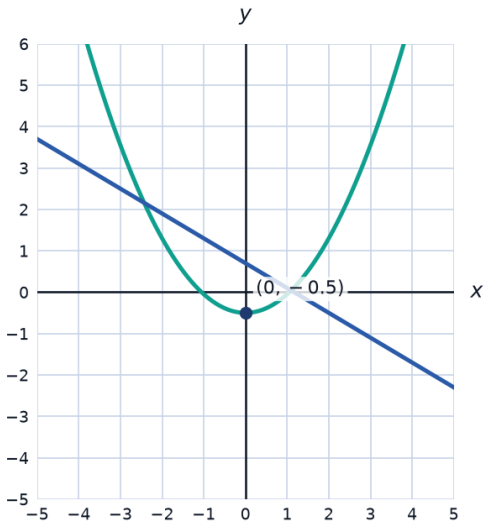
b. $[-1, 1]$

c. $(-1, 1]$

d. $[-1, 1)$

33 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

Reduction formula $I_n = \int_0^{\pi/2} \sin^n x\,dx$: $I_n = \frac{n-1}{n} I_{n-2}$. Find I_4:



a. $3\pi/8$

b. $3\pi/16$

c. $\pi/4$

d. $\pi/2$

34 [Intermediate] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

The Maclaurin series for $\sqrt{1+x}$ is a special case of the binomial series with $\alpha =$

a. 2

b. 1

c. $-1/2$

d. $1/2$

© ClipSAT — clipsat.org | All Rights Reserved

35 [Intermediate] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

The Maclaurin series for e^{-x^2} (substitute $-x^2$ for x in e^x series):

a. $\sum \frac{(-1)^n x^{2n}}{n!}$

b. $\sum x^{2n}/n!$

c. $\sum (-x)^n/n!$

d. $\sum x^n/n!$

36 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

Maclaurin series for $\tan^{-1} x$ at $x = 1$ gives $\pi/4 = 1 - 1/3 + 1/5 - 1/7 + \cdots$. This converges very slowly. The 10th partial sum approximates $\pi/4$ with error less than:

a. 1/11

b. 1/21

c. 1/19

d. 0.01

37 [Intermediate] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

Maclaurin series for e^x: first four terms:

a. $1 + x + x^2 + x^3$

b. $1 + x + x^2/2! + x^3/3!$

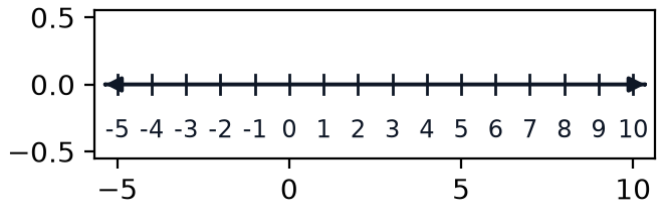
c. $x + x^2 + x^3$

d. $1 - x + x^2 - x^3$

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 5.04 Graphs of Inverse Trigonometric Functions & Their Derivatives (AA HL)

Consider the power series $\sum_{n=0}^{\infty} c_n(x-a)^n$. If the radius of convergence is $R = 3$ and the center of the series is $a = 2$, which of the following intervals represents the values of x for which the series converges?



a. $(-1, 5)$

b. $[-1, 5]$

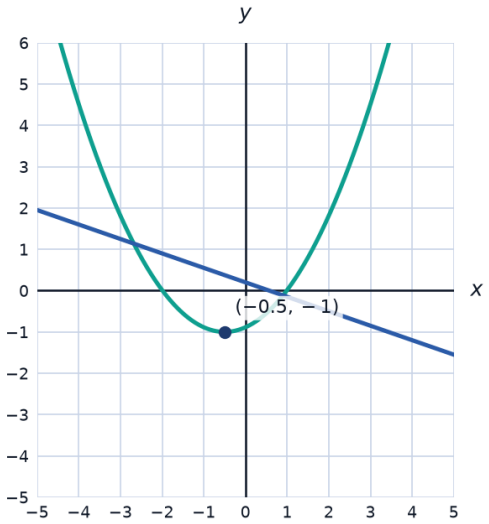
c. $(-1, 5]$

d. $[-1, 5)$

© ClipSAT — clipsat.org | All Rights Reserved

39 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 6.01 Matrices (AI HL)

A sequence satisfies $u_{n+1} = 2u_n + 4$ with $u_1 = 3$. Find u_4.



a. 59

b. 33

c. 52

d. 39

40 [Challenging] • Target: Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14) • Cross-Lesson: 6.02 Graph Theory & Networks (AI HL)

A graph has vertices $\{1, \ldots, 6\}$ and edges $1{-}5, 1{-}4, 4{-}5, 2{-}6, 2{-}5, 1{-}3$. Find the degree of vertex 2.

a. 0

b. 1

c. 3

d. 2

41 [Challenging] • Target: Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20) • Cross-Lesson: 6.03 Markov Chains (AI HL)

A sequence is defined by $u_1 = 8$ and $u_{n+1} = u_n + 6$. Find u_4 given $u_3 = 20$.

a. 26

b. 34

c. 27

d. 32

© ClipSAT — clipsat.org | All Rights Reserved

42 [Challenging] • Target: Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28) • Cross-Lesson: 1.01 Proof by Mathematical Induction

State the principal value (in degrees, $-90^\circ \le \theta \le 90^\circ$) satisfying $\sin\theta = 0$.

a. 16°

b. 0°

c. -34°

d. 18°

43 [Challenging] • Target: Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37) • Cross-Lesson: 1.02 Complex Numbers: Cartesian Form & Algebra

A graph's adjacency matrix is $A = (0,1,1;1,0,1;1,1,0)$ (vertices 1,2,3). Find the number of walks of length 2 from vertex 1 to vertex 1 (the $(1,1)$ entry of A^2).

a. 1

b. 0

c. 2

d. 4

44 [Challenging] • Target: Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45) • Cross-Lesson: 1.03 Polar Form, De Moivre's Theorem & Roots of Unity

Find the future value of an ordinary annuity of $x500 deposited annually for 6 years at 2\%$ annual interest, correct to 2 decimal places.

a. \2,838.65

b. \3,154.06

c. \3,469.47

d. \3,627.17

45 [Challenging] • Target: Free-Response Questions (FRQs): Foundational Fluency (1–7) • Cross-Lesson: 1.04 Partial Fractions

Consider the power series $\sum_{n=1}^{\infty} \frac{(2x-1)^n}{n \cdot 3^n}$. For what values of x does this series converge conditionally?

a. $-1 \le x < 2$

b. $-1 < x \le 2$

c. $-1 \le x \le 2$

d. $-1 < x < 2$

© ClipSAT — clipsat.org | All Rights Reserved